Denominator bounds and rational telescopers for symmetric zeta forms

Abstract

We derive a uniform denominator bound for the normalized symmetric kernels, construct their rational telescoping certificates after lower-period cancellation, and retain the full gcd in the resulting arithmetic cost. The denominator theorem is a paper proof. Lean verifies the leading binomial identity, block recurrence, positivity implication, and invariance under excess clearing. No all-odd irrationality theorem is claimed.

1 An explicit denominator budget

Fix an odd $s = 2r + 1 \geq 3$, an even $n \geq 0$, and $0 \leq m \leq n$. Define

$$T_{n,m}(t) = \frac{(n!)^s}{(m!)^2} \frac{\prod_{v=1}^{m}(t-v)(t+n+v)}{\prod_{j=0}^{n}(t+j)^s}, \qquad d_n = \operatorname{lcm}(1,\ldots,n), \quad d_0 = 1.$$

This proper rational function has a unique expansion

$$T_{n,m}(t) = \sum_{j=0}^{n} \sum_{q=1}^{s} \frac{c_{j,q}}{(t+j)^q}.$$

Proposition 1. *For every displayed j, q,*

$$d_n^{s-q} c_{j,q} \in \mathbb{Z}. \tag{1}$$

Consequently the convergent sum has the form

$$\mathcal{A}_{n,m} := \sum_{k \geq 1} T_{n,m}(k) = b_{n,m} + \sum_{\substack{3 \leq q \leq s \\ q \text{ odd}}} a_{q,n,m} \zeta(q),$$

where $d_n^{s-q} a_{q,n,m} \in \mathbb{Z}$ and $d_n^s b_{n,m} \in \mathbb{Z}$.

Proof. Factor $T_{n,m} = P_m Q_n^s$, where

$$P_m(t) = \binom{t-1}{m}\binom{t+n+m}{m}, \qquad Q_n(t) = \frac{n!}{\prod_{h=0}^{n}(t+h)}.$$

For any integer a, Vandermonde's polynomial identity and

$$\binom{a+x}{m} = \sum_{i=0}^{m} \binom{a}{m-i}\binom{x}{i}, \qquad \binom{x}{i} = \frac{(-1)^{i-1}x}{i} \prod_{h=1}^{i-1}(1 - x/h) \quad (i \geq 1)$$

show that d_m^ℓ clears its coefficient of x^ℓ. Indeed each contributing denominator is a product of ℓ integers from $1, \ldots, i$. The coefficients $\binom{a}{k}$ are integral also for negative a, by $\binom{-b}{k} = (-1)^k \binom{b+k-1}{k}$. Products retain the weighted Taylor bound. Since $m \leq n$, it holds with d_n for $P_m(-j + x)$.

At a pole,

$$[xQ_n(-j+x)]^s = (-1)^{js}\binom{n}{j}^s \prod_{h\neq j}(1+x/(h-j))^{-s}.$$

The coefficient of x^ℓ in each factor is $(-1)^\ell\binom{s+\ell-1}{\ell}(h-j)^{-\ell}$; d_n^ℓ clears it. Thus the coefficient of x^{s-q} in $P_m(-j+x)[xQ_n(-j+x)]^s$ satisfies (1). That coefficient is exactly $c_{j,q}$.

Reflection gives $c_{n-j,q} = (-1)^{q+1}c_{j,q}$, canceling even zeta coefficients. Proper decay gives $\sum_j c_{j,1} = 0$, and partial-fraction summation yields

$$a_{q,n,m} = \sum_j c_{j,q}, \qquad b_{n,m} = -\sum_{j=0}^{n}\sum_{q=1}^{s} c_{j,q}H_j^{(q)}.$$

For $q = 1$, subtract $H_K\sum_j c_{j,1} = 0$ from the K-th partial sum; the remaining upper boundary tends to zero since $0 \le H_{K+j} - H_K \le n/(K+1)$. No value of $\zeta(1)$ is used. Finally $d_n^q H_j^{(q)} \in \mathbb{Z}$, proving the assertion about $b_{n,m}$. The case $n = 0$ follows directly from $T_{0,0}(t) = t^{-s}$. □

This is an explicit application to our kernels of the integral-valued-polynomial arithmetic appearing in [2, Section 2]. It reduces the clearing range from $n+m$ to n. In particular,

$$c_{j,s} = (-1)^{m+j}\binom{n}{j}^s\binom{j+m}{m}\binom{n-j+m}{m}.$$

Lean verifies the factorial-to-binomial magnitude identity and integrality. Its identification with the Laurent coefficient is part of the paper proof.

2 The combined form and its actual arithmetic cost

Use the r blocks $m = n-j$, $0 \le j < r$, for sufficiently large even n. Choose nonzero integer weights w_j canceling the $r-1$ lower odd periods. Put $M_n = \sum_j |w_j|$ and

$$J_n = d_n^s\sum_j w_j\mathcal{A}_{n,n-j} = A_n\zeta(s) + B_n, \qquad h_n = \gcd(|A_n|, |B_n|).$$

When $h_n > 0$, the primitive form is J_n/h_n, with the bound

$$\left|\frac{J_n}{h_n}\right| \le K_n \max_j \mathcal{A}_{n,n-j}, \qquad K_n = \frac{d_n^s M_n}{h_n}. \tag{2}$$

This keeps the cancellation weights and the complete gcd in the same calculation. Replacing the clearing factor by $c\,d_n^s$ multiplies A_n, B_n, h_n by c, so K_n is unchanged. Lean verifies this invariance: sharper clearing can improve an upper bound that ignores the gcd, but cannot alter an already primitive form.

The companion note's saddle theorem [1, Section 5] gives, for every fixed j,

$$\frac{1}{n}\log\mathcal{A}_{n,n-j} \longrightarrow \lambda_s < 0 \quad (n\longrightarrow\infty,\ n \text{ even}).$$

For $s = 5$, $\lambda_5 \in (-4.970794, -4.970793)$. Hence a sufficient arithmetic estimate would be

$$\limsup_{n\to\infty,\ n \text{ even}} \frac{\log K_n}{n} < -\lambda_s.$$

This strict inequality is not established here. Nonvanishing is also required. At positive integers $T_{n,m}(k) = 0$ for $k \le m$ and $T_{n,m}(k) > 0$ thereafter. Therefore weights of one sign give a nonzero combined sum. Lean proves the corresponding convergent-series implication; the existence of one-sign nullvectors at all indices is not proved.

3 A rational telescoping certificate

Write any combined kernel after cancellation as $F(t) = \sum_{j,q} c_{j,q}(t+j)^{-q}$, so that $\sum_j c_{j,q} = a$ for $q = s$ and zero otherwise. Using the Kronecker delta, define

$$\beta_{j,q} = \sum_{h=0}^{j} c_{h,q} - a\,\delta_{q,s}, \qquad G(t) = \sum_{j=0}^{n-1} \sum_{q=1}^{s} \frac{\beta_{j,q}}{(t+j)^q}.$$

Then the exact certificate is

$$F(t) = \frac{a}{t^s} + G(t) - G(t+1), \qquad \sum_{k\geq 1} F(k) = a\zeta(s) + G(1). \tag{3}$$

At the first pole the recovered coefficient is $a\delta_{q,s} + \beta_{0,q}$; at an interior pole it is $\beta_{j,q} - \beta_{j-1,q}$; at the last it is $-\beta_{n-1,q}$. These are exactly the $c_{j,q}$. The rational difference has no poles and vanishes at infinity, proving the first identity. Telescoping and $G(k) \to 0$ prove the second. This identifies the rational constant constructively; it does not itself improve its arithmetic.

The actual block deformation also has the exact recurrence

$$T_{n,m+1}(t) = \frac{(t-m-1)(t+n+m+1)}{(m+1)^2}\, T_{n,m}(t).$$

Lean verifies the unnormalized kernel version. These are identities in the summation variable and block parameter, not a new recurrence in n.

Verified scope. The exact audit checks all $m \leq n$ at weights $3, 5, 7, 9$ and $n = 0, 2, 4, 6, 8, 12, 20$: 236 kernels and 18 600 coefficient assertions. It also certifies 19 combined-form ledgers, all with one-sign weights, including every pole coefficient of their telescopers and the equality of $G(1)$ with the rational constant. Changing the clearing range leaves their primitive costs exactly unchanged. Five new Lean foundation reports use only standard foundations, with no added axioms or placeholders. The general denominator theorem and telescoper derivation are paper proofs.

These finite certificates do not prove an asymptotic cost bound. For example, the weight-five ledger with $m = n, n-1$ has $\log K_n/n \approx 7.286432$ at $n = 20$, compared with the limiting analytic decay rate $-\lambda_5 \approx 4.970794$. This is a finite comparison, not an impossibility theorem. The winning inequality in (2) remains unproved, so no all-odd irrationality conclusion follows.

References

[1] *Explicit symmetric Padé kernels and exact odd-zeta cancellation*, companion research note, Section 5. https://github.com/ranbirsurii/irrationality-perhaps-/blob/74629b3/notes/symmetric-pade-note.pdf.

[2] W. Zudilin, *Arithmetic of linear forms involving odd zeta values*, Journal de Théorie des Nombres de Bordeaux 16 (2004), 251–291. https://www.numdam.org/item/JTNB_2004__16_1_251_0.pdf.